

UBT512: STRUCTURAL BIOLOGY

L	T	P	Cr
3	0	0	3.0

Course Objective: Aiming to provide basic knowledge of structural components in biological systems, this course will make students aware of the lower to higher-order biological organizations, including the molecular to macromolecular structures and biological assemblies that drive the living systems.

Syllabus

Structural components of biological system: Levels of molecular organization, Brief discussions on: Amino acids (R groups and conformations, D- L- isomers), Nucleic acids (general characteristics of nucleic acid structures, forces and stabilizing geometries, glycosidic bond, rotational isomers), Carbohydrates, Lipids, Cofactors, Vitamins, and Hormones; Basic forces and interaction holding biological structure (ionic, polar, hydrogen bonding; hydrophobic, etc.); symmetry in biological molecules: translation, rotation, reflection, and glide; organization of molecules in lattice.

Protein and DNA structures: Protein structure and hierarchy; secondary structure: α -helix, β sheet, loop; $\phi/\psi/\chi$ angles: Ramachandran Plot; Super-secondary and tertiary structure: motifs, domain, and fold; Quaternary structure: oligomer assembly- protomer and oligomer: homo-oligomer and heterooligomer; biological implications of higher order protein structure - cooperative interactions, allosteric effects; structural families and classes; Structure of IgG, fibrous proteins (structure of collagen, keratin); Stabilizing ordered forms of DNA (A, B and Z), base-pairing types, base stacking, tertiary structure of DNA (Supercoiled DNA); Concept of DNA lattice; Ribose puckering and tertiary structure of tRNA.

Biological membrane and higher order structural organization: Biological membrane components-lipids and membrane proteins; properties of biological membranes; Singer Nicolson model; Membrane protein structures, function, and assemblies; lipid rafts and micro-domains; Present understanding of the structure of biological membrane and cytoskeleton; Actin; Plant plasmodesmata structure; virus structures; SNARE complexes.

Tools for determination of biological structures: Overview of the major tools to study the molecular structures, X-ray diffraction; NMR and Electron microscopy; Atomic force microscopy (AFM).

Laboratory Work (if applicable)

Course Learning Objectives (CLO)

The students will be able to:

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

1. Explain the structural components of biological system.
2. Understand the aspects of protein and DNA structure and its importance in living system.
3. Comprehend the concept of lower to higher order structural organization of life.
4. Apply the basic methods of determining and studying biological structures.

Text Books

1. Alberts B, et al., *Molecular Biology of The Cell* 7th edition, Garland Science (2022).
2. Lodish et al., *Molecular Cell Biology* 8th edition, Freeman Macmillan (2016).

Reference Books

1. Branden CI, Tooze J: Introduction to Protein Structure, Garland Science (1998).
2. Williamson M: How Proteins Work, Garland Science (2012).
3. Miller and Tanner, Essentials of Chemical Biology: Structure and Dynamics of Biological Macromolecules, Wiley, (2008).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/quizzes)	30

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